

Introduction to Contract Theory and Auctions
Bayesian Auctions: Symmetric, Independent, Private Values

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The symmetric model with independent private values

We will now study a class of models that satisfy the following assumptions:

- A1 Single object for sale and the object is indivisible.
- A2 The seller is willing to sell the object at any nonnegative price.
- A3 There are n buyers.
- A4 Private values and symmetric values: All buyers have the same set of possible private values $\mathcal{V}$. This set can be a closed bounded interval $[0, \bar{v}]$ or the set of nonnegative numbers $[0, \infty)$. Every buyer knows her private value for the object. The random values $V_1, V_2, \dots, V_n$ of the private values of the buyers are independent and identically distributed. Denote by F the common cumulative distribution function of the random variables V_i , $i = 1, \dots, n$. The support of this distribution is $\mathcal{V}$. The joint distribution is then $F^n = F \times \dots \times F$.
- A5 Continuity: For each i , the random variable V_i is continuous, and its density function, denoted by f , is continuous and positive.
- A6 Risk neutrality: All buyers are risk neutral, and therefore seek to maximize their expected profits.

A1–A6 are common knowledge among the buyers. Any auction satisfying A1–A6 is called a **symmetric auction with independent private values (SIPV)**.

Definition 6.1

In a symmetric auction with independent private values, an equilibrium profile $(\beta_1^*, \beta_2^*, \dots, \beta_n^*)$ is called a **symmetric equilibrium** if $\beta_i^* = \beta_j^*$ for all $i, j \leq n$; that is, all buyers implement the same strategy.

When the profile $\beta^* = (\beta_i^*)_{i \in N}$ is a symmetric equilibrium, we *abuse notation* and denote the common strategy also as β^* , i.e., $\beta^* = \beta_i^*$ for every $i \in N$. Such a strategy is called a **symmetric equilibrium strategy**.

Definition 6.2

A symmetric equilibrium strategy β^* is monotonically increasing if the higher the private value, the higher the buyer's bid:

$$v < v' \Rightarrow \beta^*(v) < \beta^*(v') \quad \forall v, v' \in \mathcal{V}.$$

Example: Two bidders with i.i.d Uniform valuations

Example 6.3

Suppose that V_1 and V_2 are independent and identically distributed (i.i.d.) Uniform random variables over the interval $[0, 1]$, i.e., $V_1, V_2 \sim U[0, 1]$. In this example, we will show that in a sealed-bid first-price auction, the following strategy is a symmetric equilibrium:

$$\beta_i^*(v_i) = \frac{v_i}{2} \quad i = 1, 2.$$

Theorem 6.4

Let $Y = \max_{j \neq i} \{V_j\}$ be the random variable whose value equals the highest private value of all buyers j other than i , and let G and g denote respectively the CDF and PDF of Y . In a symmetric first-price auction with independent private values the following strategy defines a monotonically increasing, continuous, differentiable, symmetric equilibrium:

$$\beta(v) = E[Y|Y \leq v], \quad \forall v \in \mathcal{V} \setminus \{0\}$$

$\downarrow$
 $[0, \bar{v})$
 $[0, \infty)$

and $\beta(0) = 0$.

Proof.

Step 1: $\beta(v)$ is monotonically increasing.

Remark: For any R.V. X and every pair of events such that $A \cap B = \emptyset$

$$E[X|A \cup B] = P(A|A \cup B) E[X|A] + P(B|A \cup B) E[X|B]$$

Note that $g > 0$ for all y .

Let v be an interior point of $\mathcal{V} = [0, \bar{v}]$, $\Rightarrow v > 0, v < \bar{v}$.

Then for any $\delta > 0$ such that $v + \delta < \bar{v}$ we want to show that

$$\beta(v + \delta) > \beta(v) \quad \text{Goal!}$$

$$\beta(v + \delta) = E[Y | Y \leq v + \delta] \quad \text{We decompose } Y \leq v + \delta \text{ as two events}$$

$$Y \leq v \quad \text{AND} \quad v < Y \leq v + \delta$$

$$= \Pr(Y \leq v | Y \leq v + \delta) E[Y | Y \leq v]$$

$$+ \Pr(v < Y \leq v + \delta | Y \leq v + \delta) E[Y | v < Y \leq v + \delta]$$

$$= \frac{\Pr(Y \leq v)}{\Pr(Y \leq v + \delta)} E[Y | Y \leq v] + \frac{\Pr(v < Y \leq v + \delta)}{\Pr(Y \leq v + \delta)} E[Y | v < Y \leq v + \delta]$$

Since g is positive: $0 < E[Y | Y \leq v] \leq v < E[Y | v < Y \leq v + \delta]$ because $v > 0$

$\beta(v + \delta)$ is a weighted average of positive terms and $\beta(v) \leq v$

$$\Rightarrow \beta(v + \delta) > \beta(v)$$

Heuristic derivation of the symmetric eq. strategy with n bidders in the SIPV First-Price Auction.

Suppose that bidders $j \neq 1$ follow the symmetric, monotonically increasing, and differentiable equilibrium strategy β . Suppose that bidder 1 has value $V_1 = v_1$ and bids b .

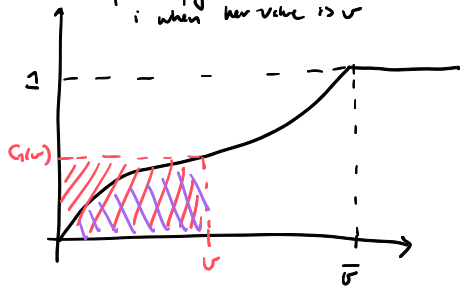
$$\begin{aligned}\beta(v_i) &= E[Y \mid Y \leq v] \quad ; \quad Y = \max_{j \neq i} V_j \sim G(y) \text{ ; pdf } g(y) \\ &= \frac{1}{G(v)} \int_0^v y g(y) dy \quad \quad \quad = Y_1^{(n-1)} \sim G_1^{(n-1)}(y) \text{ ; } g_1^{(n-1)}(y)\end{aligned}$$

Note that $\beta(v) G(v) = \int_0^v y g(y) dy = y G(y) \Big|_0^v - \int_0^v G(y) dy = v G(v) - \int_0^v G(y) dy$

$u = y \quad s = G(y)$

$du = dy \quad ds = g(y) dy$

$\therefore \underbrace{\beta(v) G(v)}_{\text{expected payment of player } i \text{ when her value is } v} = v G(v) - \underbrace{\int_0^v G(y) dy}_{\text{degree of value shading in FPA}}$



$G(v)$

$$\boxed{\beta(v) = v - \int_0^v \frac{G(y)}{G(v)} dy}$$

Proof (Cont.)

Step 2: $\beta(v)$ is continuous.

Continuity at $v=0$, For each $v>0$: $0 \leq \beta(v) = E[Y|Y \leq v] \leq v$.

Taking the limit $v \rightarrow 0 \Rightarrow 0 \leq \beta(0) \leq 0 \Rightarrow \beta(0) = 0$.

Continuity at interior points of $\mathcal{V}$

$\beta(v) = \frac{\int_0^v y g(y) dy}{G(v)}$, we have a quotient of two continuous functions and $G(v) \neq 0$ in $v \in (0, \bar{v}) \Rightarrow \beta(v)$ is continuous

Step 3: $\beta(v)$ is differentiable. $\beta(v) = v - \int_0^v \frac{G(y) dy}{G(v)}$

$$\beta'(v) = 1 - \frac{G^2(v) - g(v) \int_0^v G(y) dy}{G^2(v)} = \left(\frac{g(v)}{G(v)} \right) \left(\int_0^v \frac{G(y)}{G(v)} dy \right)$$

Actually β is continuously differentiable.

Proof (Cont.)

Step 4: $\beta(v)$ is a symmetric equilibrium. Suppose that all bidders other than i follow β
Goal: show that it is a best response for i to bid according to β given her value v_i



Theorem 6.5

In the symmetric equilibrium given by Theorem 6.4, the expected payment that a buyer with private value v makes for the object is $G_Y(v) \times E[Y|Y \leq v]$ in both sealed-bid first-price and second-price auctions.

Proof.



Theorem 6.6

In a symmetric sealed-bid auction with independent private values, at the symmetric equilibrium, the expected revenue of a seller in both sealed-bid first-price and second-price auctions, is

$$\pi = n \int_{v \in \mathcal{V}} G_Y(v) \mathbb{E}[Y | Y \leq v] f(v) dv$$

Proof.

